

Faculty of Computing, Electronics
and Science

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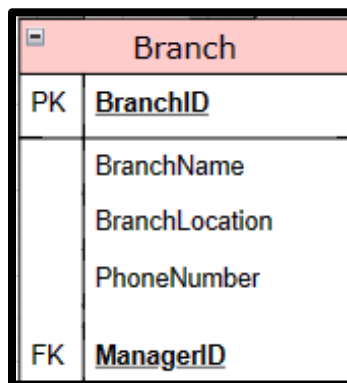
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ENTITY-RELATIONSHIP DIAGRAM (ERD)

ENTITIES:

1. Branch

- Attributes:
 - BranchID (Primary Key)
 - BranchName
 - BranchLocation
 - PhoneNumber



- Relationships:
 - Manages (1:N) with Staff (FK: ManagerID)

2. Staff

- Attributes:
 - StaffID (Primary Key)
 - Name

- Position
- Email

Staff	
PK	<u>StaffID</u>
FK	<u>PropertyCode</u>
FK	<u>BranchID</u>
	Name
	Position
	Email

- Relationships:
 - WorksAt (N:1) with Branch (FK: BranchID)
 - Manages (0..1:1) with Branch (FK: BranchID)
 - AssignedTo (1:N) with RentalProperties (FK: PropertyCode)

3. Manager

- Attributes:
 - ManagerID (Primary Key)
 - Name
 - Email

Manager	
PK	<u>ManagerID</u>
	Name
	Email
FK	<u>BranchID</u>

- Relationships:
 - Manages (1:1) with Branch (FK: BranchID)

4. RentalProperty

- Attributes:
 - PropertyCode (Primary Key)

- Area
- Address
- Bedrooms
- Bathrooms
- ReceptionRooms
- Garage
- PropertyType
- MonthlyRental

RentalProperty	
PK	<u>PropertyCode</u>
FK	<u>StaffID</u>
FK	<u>ClientID</u>
	Area
	Address
	Bedrooms
	Bathrooms
	ReceptionRooms
	Garage
	PropertyType
	MonthlyRental

- Relationships:
 - AssignedTo (1:N) with Staff (FK: StaffID)
 - RentedBy (0..1:1) with RentalClients (FK: ClientID)

5. RentalClient

- Attributes:
 - ClientID (Primary Key)
 - Name
 - Email
 - ContactNumber

RentalClient	
PK	<u>ClientID</u>
FK	<u>PropertyCode</u>
	Name
	Email
	ContactNumber

- Relationships:
 - Rents (0..1:N) with RentalProperties (FK: PropertyCode)

6. PropertySeller

- Attributes:
 - SellerID (Primary Key)
 - Name
 - Email
 - ContactNumber

PropertySeller	
PK	<u>SellerID</u>
FK	<u>PropertyCode</u>
	Name
	Email
	ContactNumber

- Relationships:
 - Sells (1:N) with PropertiesToBeSold (FK: PropertyCode)

7. PropertiesToBeSold

- Attributes:
 - PropertyCode (Primary Key)
 - Area
 - Address
 - Bedrooms

- Bathrooms
- ReceptionRooms
- Garage
- PropertyType
- FreeholdOrLeasehold
- Price
- CurrentState
- DateOnMarket

PropertyToBeSold	
PK	<u>PropertyCode</u>
FK	BuyerNo
	Area
	Address
	Bedrooms
	Bathrooms
	ReceptionRooms
	Garage
	PropertyType
	FreeHoldOrLeaseHold
	Price
	CurrentState
	DateOnMarket

- Relationships:
 - InterestedBuyers (1:N) with ProspectiveBuyers (FK: BuyerNo)

8. ProspectiveBuyers

- Attributes:
 - BuyerNo (Primary Key)
 - Name
 - Email
 - ContactNumber
 - MinBedrooms
 - MinBathrooms

- MinReceptionRooms
- GarageRequired
- MaxPrice

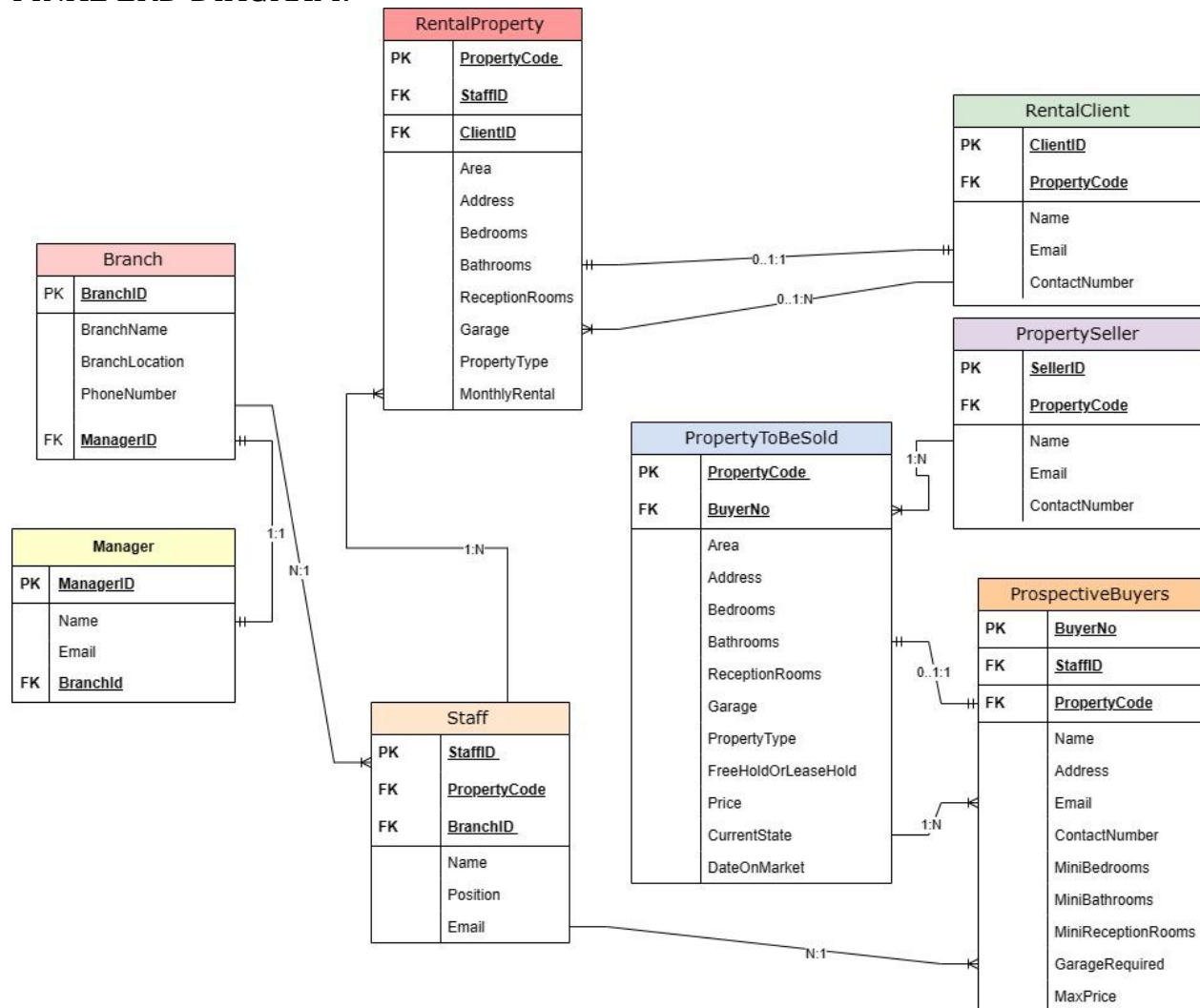
ProspectiveBuyers	
PK	BuyerNo
FK	StaffID
FK	PropertyCode
	Name
	Address
	Email
	ContactNumber
	MiniBedrooms
	MiniBathrooms
	MiniReceptionRooms
	GarageRequired
	MaxPrice

- Relationships:
 - MakesAppointmentWith (N:1) with Staff (FK: StaffID)
 - MakesOffer (0..1:1) with PropertiesToBeSold (FK: PropertyCode)

RELATIONSHIPS:

- **Manages:** Connects Manager to Branch (1:1)
- **WorksAt:** Connects Staff to Branch (N:1)
- **AssignedTo:** Connects Staff to RentalProperty (1:N)
- **RentedBy:** Connects RentalProperty to RentalClient (0..1:1)
- **Rents:** Connects RentalClient to RentalProperty (0..1:N)
- **Sells:** Connects PropertySeller to PropertiesToBeSold (1:N)
- **InterestedBuyers:** Connects PropertiesToBeSold to ProspectiveBuyers (1:N)
- **MakesAppointmentWith:** Connects ProspectiveBuyers to Staff (N:1)
- **MakesOffer:** Connects ProspectiveBuyers to PropertiesToBeSold (0..1:1)

FINAL ERD DIAGRAM:



PRIMARY KEYS (PK) AND FOREIGN KEYS (FK):

- **Branch:**
 - PK: BranchID
- **Staff:**
 - PK: StaffID
 - FK: WorksAt (references Branch)
 - FK: Manages (references Branch)
- **Manager:**
 - PK: ManagerID

- FK: Manages (references Branch)
- **RentalProperty:**
 - PK: PropertyCode
 - FK: AssignedTo (references Staff)
 - FK: RentedBy (references RentalClient)
- **RentalClient:**
 - PK: ClientID
- **PropertySeller:**
 - PK: SellerID
- **PropertiesToBeSold:**
 - PK: PropertyCode
 - FK: Sells (references PropertySeller)
- **ProspectiveBuyers:**
 - PK: BuyerNo
 - FK: MakesAppointmentWith (references Staff)
 - FK: MakesOffer (references PropertiesToBeSold)

ASSUMPTIONS:

1. **Branch:**
 - Assumption: The branch location is considered as a general identifier, and detailed address information is stored in the "Address" attribute.
 - Assumption: Phone numbers are stored as strings to accommodate various formats.
2. **Staff:**
 - Assumption: Date of birth is used for age calculation and is stored in the "DateOfBirth" attribute.
 - Assumption: Email and phone are unique identifiers for staff members.
3. **Manager:**
 - Assumption: Each branch has one manager, and the manager is uniquely identified by their ManagerID.

- Assumption: Email and phone are unique identifiers for managers.

4. **RentalProperty:**

- Assumption: AvailabilityStatus indicates whether the property is currently available for rent.
- Assumption: LeaseDuration represents the maximum lease period in months.
- Assumption: Boolean attributes like AllowsPets and AllowsChildren are used to capture client preferences.

5. **RentalClient:**

- Assumption: EmploymentStatus records the current employment status of the rental client.
- Assumption: RentalHistory provides a summary of the client's past rental experiences.

6. **PropertySeller:**

- Assumption: PropertiesSold represents the total number of properties the seller has sold historically.
- Assumption: Email and phone are unique identifiers for property sellers.

7. **PropertiesToBeSold:**

- Assumption: FreeholdOrLeasehold indicates whether the property is freehold or leasehold.
- Assumption: CurrentState represents the current status of the property in the selling process.
- Assumption: DateOnMarket indicates when the property was listed for sale.

8. **ProspectiveBuyers:**

- Assumption: PreferredArea specifies the preferred location or neighborhood for prospective buyers.
- Assumption: Boolean attribute GarageRequired signifies whether the buyer requires a property with a garage.
- Assumption: Email and phone are unique identifiers for prospective buyers.

9. **Relationships:**

- Assumption: N:1 relationships (e.g., WorksAt, Manages) indicate that multiple entities on the "N" side can be associated with one entity on the "1" side.
- Assumption: 0..1:1 relationships (e.g., RentedBy, MakesOffer) imply optional associations.

10. General:

- Assumption: The ERD focuses on essential relationships and attributes, and additional details not explicitly mentioned in the scenario are excluded for simplicity.
- Assumption: The ERD design assumes that the provided scenario comprehensively captures the essential aspects of Peter Darwin Estate Agency's operations.

These assumptions aim to clarify specific decisions made during the design of the ERD, balancing between simplicity and completeness.